Network Analysis of Character Relationships in Harry Potter Books

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Abstract

In this project, we construct and analyze a character co-occurrence network from the Harry Potter book series. Using paragraph-level co-appearances as a proxy for interaction strength, we build a weighted, undirected network for each individual book as well as for the entire saga. We examine key structural properties of these networks—including degree distributions, density, clustering, path lengths, and centrality measures—to characterize the roles and importance of individual characters and understand the topology. Community detection methods, such as the Louvain algorithm and clique-percolation, are applied to reveal underlying social groupings and narrative substructures. After analyzing community structure, we also investigate several global phenomena, including the small-world property, robustness to random failure and targeted attacks, and the temporal evolution of the network across the seven books. Finally, we explore different generative models that could explain the observed network topology. Overall, our results show that the social network in the Harry Potter series is well described by the Bianconi–Barabási: relationship formation in the narrative follows growth and preferential attachment with fitness, where central characters emerge as hubs due to their narrative importance while others remain peripheral—consistent with a scale-free topology. These findings demonstrate that network science can yield meaningful insights into fictional worlds.

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1 Introduction and motivation

The motivation behind this project was to explore the network formed by the characters in the best-seller book series Harry Potter, written by J. K. Rowling. After learning about real world networks in this course, including power-line, biologic, digital and social networks, we were curious to see if the network formed by the characters in a fictional story would reflect and exhibit the same characteristics; whether it would behave similarly to real-world social networks or if it would manifest a different behaviour altogether. Given the nature of storytelling, where some characters naturally dominate interactions while others remain peripheral, we hypothesized that the network might exhibit scale-free properties. We chose this particular book series because of its length (seven books) and its wide range of characters, think it would have a high chance of resembling real-world connections, since the story takes place in a ficitional school where social relationships and character dynamics are very important.

The other reason for choosing this topic was to see if we could apply the contents of this course in a practical and innovative way by providing insight into the field of storytelling, which traditionally hasn't been studied through the lens of network science.

Some previous approaches have attempted to model the character network of Harry Potter. However, these studies were largely aimed at the fandom and had limitations: networks were often constructed from secondary sources such as films or fanpage wikis, or they considered only a reduced subset of characters. In contrast, our goal is to investigate whether there are underlying patterns that govern the development of social networks in books and to explore how such insights could serve as a practical tool for professionals in literature, storytelling, or related fields.

2 Description of data and preprocessing

2.1 Description of chosen system and dataset

We used to different information sources to generate to create the list of nodes and edges and compute the final network: an official fan website [3] from which we obtained a CSV file containing the names, descriptions and categories (represented as images) of the characters in the Harry Potter book series via webscrapping, and full texts for the full book series from Kaggle [4].

In order to obtain the nodes, we extracted all the names and characters and assigned a unique name to each of them. We made sure to include all of the different ways a character could be referred to throughout the books and grouped them all under the same unique name identifier. To create the links between the nodes, we applied a three-paragraph sliding window and counted the number of times two characters appeared in different chapters. Isolated character appearances were also taken into account. We chose this window size due to we find it reasonable after analyzing the nature of the narrative and dialogues.

2.2 Definition of nodes and edges

The nodes in the network represent each of the characters, represented by a unique name or nickname identifier (depending on each case) and taking into consideration the cases where two characters hold the same first name.

The edges in the network are weighted links that represent the number of times the linked characters appear together in different chapters in the books, considering the three-step sliding window described in section 2.1.

2.3 Data cleaning, filtering and transformation steps

The data preprocessing consisted of several steps. The character names and descriptions obtained via web-scrapping were not ready to input into the network directly, because there were some inconsistencies, duplicated rows, mixed names and descriptions and non alpha-numerical characters (", ., ?, -) mixed in with the names which would have make the link retrieval process difficult and not representative of the real connections in the books.

The first step was to look for null values in the dataset, input the missing data in the rows where the name and description were both in the same column, and delete the ones where the information was truly missing.

There were two cases: all the information being in the "full name" column, in which case we extracted the description and added it to the "description" column; and the opposite, in which case we had to extract and fill in the "full name" column. In order to differentiate between the two we split the text strings with using the "-" character.

After splitting the information and filling in the dataset, we analyzed the "full name" column and removed any unnecessary characters that were present (',' '(' "' ')' '.' '?' '“' "”' '-'). Some of them were just noise from the encoding configuration in the original website, others were separating the names from nicknames, so we extracted them into a new "nickname" column and processed them separately.

Another issue that arose was that some rows contained multiple characters. We identified them by searching for the "and" stopword and split them into a single character per row.

Finally, we ordered the "full name" column following a *Name Surname* format and extracted the name (or name surname for duplicated first names) to create a unique identifier for each character. We also manually included additional rows with the most common names for each character (name, full name, nicknames) to ensure the edge list was as accurate as possible.

2.4 Network visualization

In Figure 2.1, we can observe a representation of the full network, containing the relations across all books. The network was visualized using NetworkX's spring layout, which arranges nodes in a way that minimizes overlap and provides more space between them for clarity. The sizes of the nodes are proportional to their degree, which measures the number of direct connections a character has with others. The opacity of the edges is proportional to the logarithm of their weight, representing the strength of interactions.

From the visualization, we can see that there are some central hubs corresponding to the main characters—such as Harry, Ron, and Hermione—who have high degrees and high betweenness centrality. These characters serve as bridges connecting different parts of the network, reflecting their narrative significance. At the same time, there are many smaller-degree nodes representing secondary characters. Interestingly, the network is quite interconnected: even at the periphery, these secondary characters form clusters or communities, which likely correspond to subplots or tightly knit groups within the story. As expected, Harry emerges as the most important character in the entire series, as he has both the largest degree and the highest betweenness centrality, meaning he interacts with the most characters and plays a crucial role in connecting different groups within the narrative.

3 Methodology and tools used

The raw text for the *Harry Potter* series was obtained through automated web scraping. We used Python's `requests` library to retrieve chapter content and `BeautifulSoup` from the `bs4` package to parse the HTML structure. Each chapter was segmented into paragraphs, which served as the fundamental units for detecting character co-occurrences.

Global

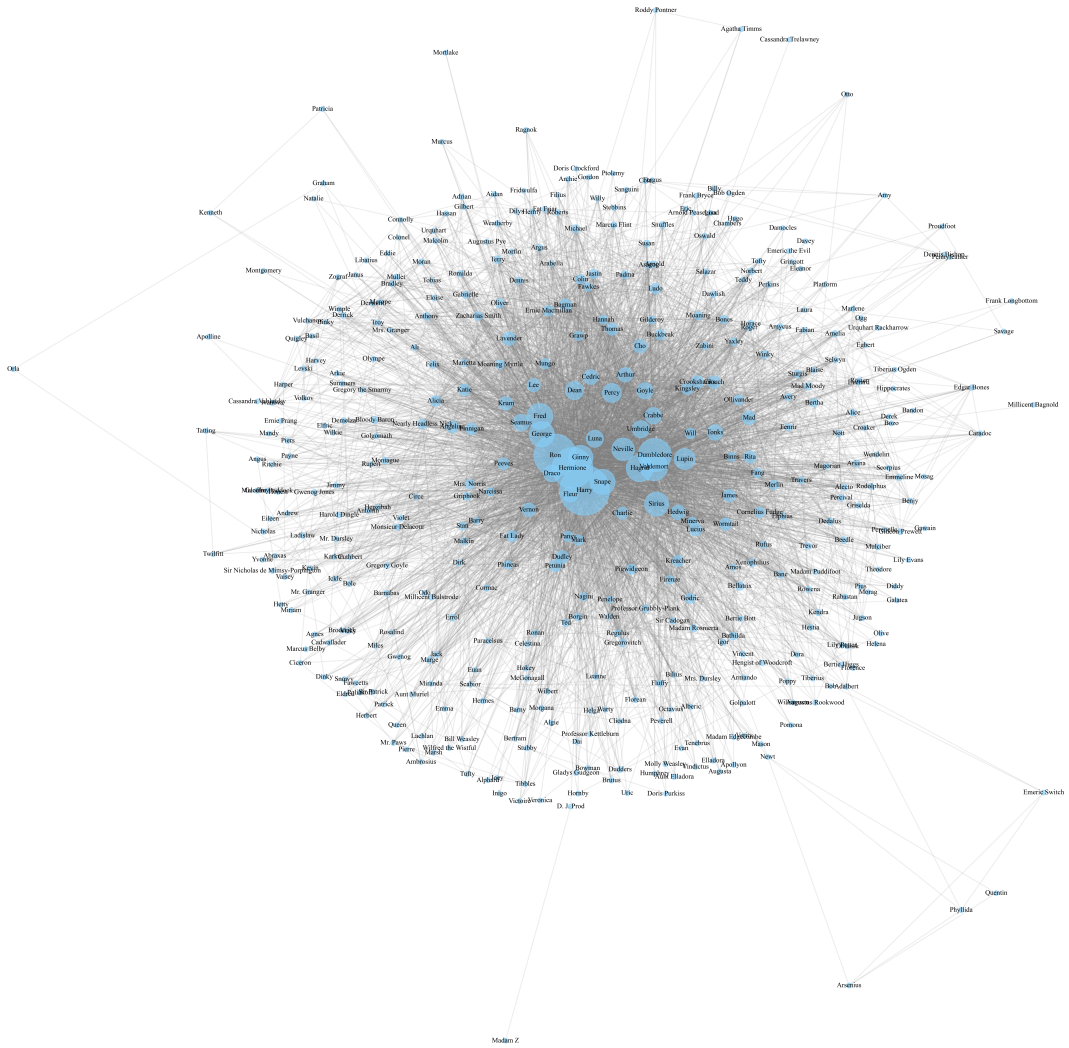


Figure 2.1: Network graph from all books.

Before constructing the network, we performed extensive cleaning of the extracted text. This included removing formatting artifacts, correcting OCR inconsistencies, and standardizing character names (e.g., mapping alternative forms such as “Professor Dumbledore” or “Albus” to a single canonical name). We built a custom dictionary of character aliases and applied rule-based matching to ensure consistent entity recognition. Paragraphs were then tokenized and scanned for all combinations of characters appearing together.

Using the cleaned data, we created a weighted, undirected graph for each book. Nodes represent characters, and an edge between two characters indicates that they appear in the same paragraph. The weight of each edge corresponds to the number of co-occurrences across the book, thus capturing the strength of their narrative relationship. Network construction and manipulation were carried out using the `networkx` library.

We analyzed each network using standard tools from network science. Key structural measures included: degree and degree centrality, clustering coefficient, network density, average shortest path length and diameter, betweenness and closeness centrality. To uncover its structure, we applied the Louvain modularity

optimization algorithm and the k-clique percolation method to detect overlapping communities. We compared results across books to study the evolution of character interactions over time. Finally, we explored different generative models to explain the observed network topology. To make these models as comparable as possible to the real networks, we estimated their parameters based on the networks' structural metrics.

All computations were performed in Python. The main libraries used were: `BeautifulSoup` and `requests` for web scraping, `pandas` for data manipulation, `networkx` for network construction and analysis, and `matplotlib` and `seaborn` for data visualization. 9 Figures and tables were generated programmatically and integrated into the final report using LaTeX.

4 Network analysis and results

4.1 Basic structural parameters

N	L	d	$\langle k \rangle$	$k_{\max}$	$k_{\min}$	$\langle C \rangle$	$\langle d \rangle$	d_{max}	r
440	5444	0.056	24.75	408	1	0.79	2.02	4	-0.33

Table 4.1: Global Harry Potter network statistics.

We can observe that the average is approximately 25, which is a reasonable number of links for a character in a school setting to have.

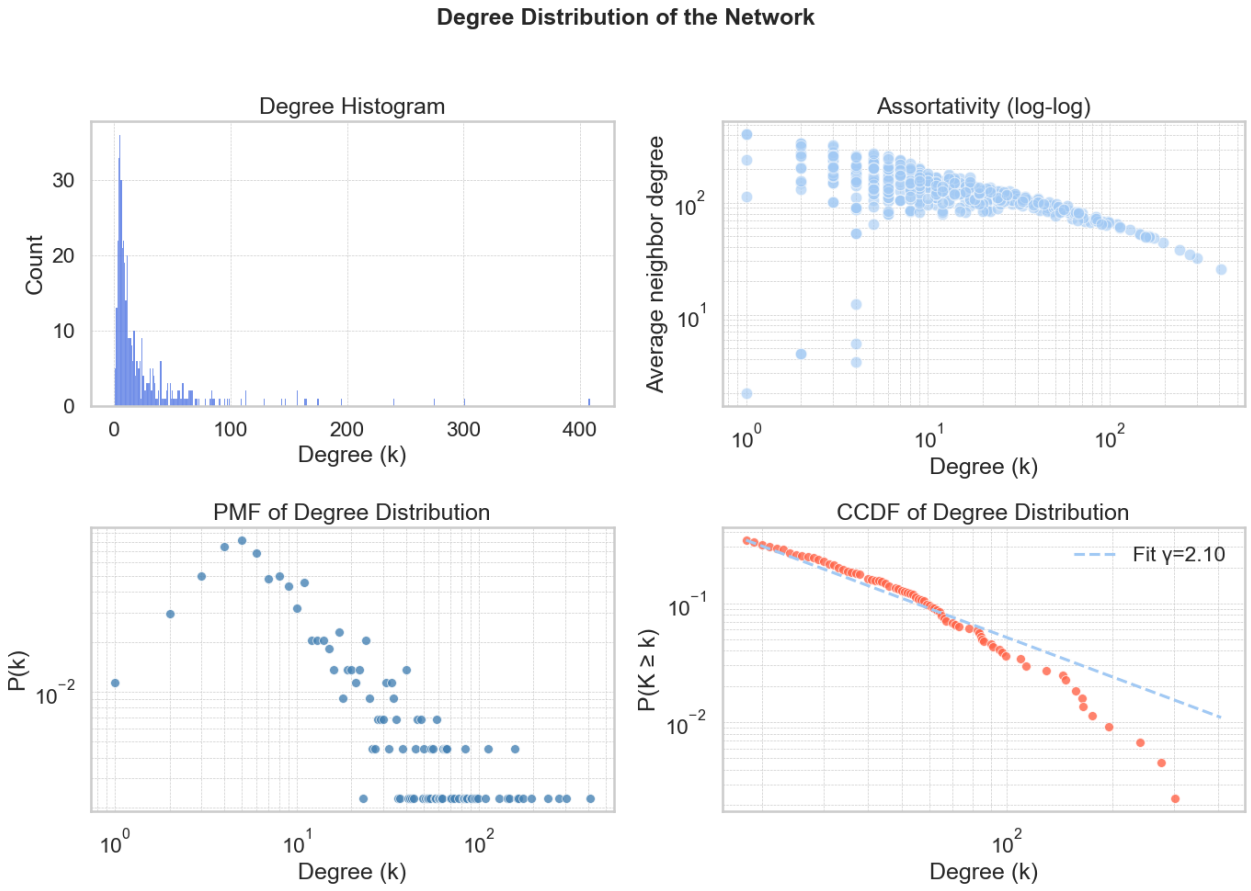


Figure 4.1: Degree distribution of the global network.

The degree CCDF of the global network displays a heavy-tailed distribution that closely follows a power law with exponent $\gamma = 2.10$, indicating the presence of highly connected hub characters. The straight-line behavior in log–log scale supports scale-free structure at intermediate degree values.

The estimated exponent $\gamma = 2.10$ is also consistent with empirical social and collaboration networks, which typically fall in the range $2 < \gamma < 3$. [1] A value close to 2 implies a particularly dense hub structure, where the highest-degree nodes accumulate a disproportionately large share of the interactions, which is consistent with the story structure where the main focus are the three protagonists, Harry, Ron and Hermione (whose nodes will be explored in more detail in later sections).

4.1.1 Degree distribution and average degree

The degree distribution shows that there are a few nodes with a very high degree (over 200) but that most of them have a lower degree, in the [0, 20] range. This is aligned with the distribution of characters and relationships in the books (or any story for that matter), since the protagonists will have a higher degree and connections to most of the other characters, while the supporting cast will have a few relationships and probably connections to the main character, but a lot less relevance overall.

The $k_{\text{nn}}(k)$ curve exhibits a clear decreasing pattern, showing that high–degree nodes tend to connect to low–degree nodes. This is typical of **disassortative mixing**, the structural signature of hub–dominated networks. The assortativity coefficient for the global network is -0.32. In the Harry Potter network, the main hubs accumulate many links from peripheral characters who do not interact strongly among themselves. This behavior is consistent with a **Bianconi–Barabási fitness model**, where nodes with higher intrinsic "fitness" attract disproportionately more connections, producing a hierarchical, core–periphery structure.

4.1.2 Clustering coefficient

The clustering coefficient is quite high. This is to be expected since most of the story revolves around a few main characters who attract most of the attention, so all the other characters will be connected to them. In a comparable random graph with density $p = 0.056$, the expected clustering would be $C_{\text{ER}} \approx p$, but the observed value is $C = 0.787$, more than an order of magnitude higher. This large discrepancy confirms strong triadic closure, typical of social networks.

4.1.3 Path length and diameter

The average path length is very close to 2, which is very short. The combination of high clustering and short path length is a manifestation of its small-world structure. This reflects the fact that most characters are connected to Harry Potter, so his node can be a bridge between almost any two other nodes in the network. The maximum degree is very close to the number of nodes in all the networks, and Harry Potter is the node with the maximum degree. Following the same logic, the diameter of the global network is 4 because even if two nodes aren't connected to the main Harry Potter node, it's highly likely that their neighbors are.

4.1.4 Density

The graphs has a density of 0.06, which means that it has a medium density (it is neither too dense or too sparse). It has enough edges for a robust connectivity, and the nodes are moderately connected. It falls in the typical density range for social networks. [5].

node	degree	local_clustering	degree_centrality	betweenness_centrality	closeness_centrality
Harry	322	0.058706	0.759434	0.408071	0.803030
Harvey	1	0.000000	0.002358	0.000000	0.445846
Dumbledore	192	0.125164	0.452830	0.084057	0.634731
Voldemort	111	0.233743	0.261792	0.018154	0.561589
Godric	26	0.523077	0.061321	0.005172	0.480726
...					

Table 4.2: Node-level metrics (degree, clustering, and centralities) for the global character network. Only the first and last five nodes are shown.

4.1.5 Centrality measures

This table reports key node-level structural measures for the global character network. Characters such as Harry, Dumbledore, and Voldemort exhibit very high degrees and centrality values, reflecting their central narrative roles and frequent interactions. In contrast, minor characters such as Harvey, Scorpius, and Hugo have extremely low degrees, zero betweenness, and low closeness, indicating peripheral positions within the network. The diversity in clustering values shows that some characters are embedded in tightly connected subgroups (clustering close to 1), while others serve as bridges between different parts of the network.

4.2 Communities

4.2.1 Non-overlapping communities

To identify cohesive groups of characters in the network, we applied several standard community detection algorithms that return *non-overlapping* partitions: the Louvain method, greedy modularity optimisation, and the Girvan–Newman algorithm. These methods are useful for obtaining a global view of the narrative structure.

Among these algorithms, Louvain produced the most coherent and interpretable partition, with a moderate modularity value indicating meaningful but not sharply separated communities. Table 4.3 summarises the structural properties of the Louvain communities in the global graph.

Community ID	Size	Internal Edges	Density	Avg. Internal Degree	Avg. External Degree
1	121	785	0.108	12.975	12.736
0	110	376	0.063	6.836	10.727
5	68	352	0.155	10.353	20.309
2	53	183	0.133	6.906	19.717
3	41	194	0.237	9.463	25.366
6	37	161	0.242	8.703	14.757
4	5	7	0.700	2.800	1.000
7	5	10	1.000	4.000	2.800

Table 4.3: Summary of Louvain communities detected in the global character network.

The detected communities exhibit strong heterogeneity in size and density. Large communities (IDs 1, 3 and 4), ranging from 72 to 151 characters, have relatively low internal densities (approximately 0.10–0.14). As community size increases, the proportion of realised edges decreases due to limited opportunities for complete mutual interaction. In contrast, several small communities (IDs 6, 10, 8 and 5) have densities close to 1.0, indicating almost fully connected subgraphs. These groups probably correspond to compact narrative units such as families, recurring side characters, or small factions.

The largest community (ID 1) has a substantial internal degree ($\langle k_{\text{int}} \rangle \approx 14.4$) but also many external connections ($\langle k_{\text{ext}} \rangle \approx 12.0$), suggesting that this group forms a major narrative core. Smaller communities, such as ID 2, combine high density with large external degree, indicating that they act as bridges between

multiple narrative modules.

The greedy modularity algorithm produced a similar number of communities but with slightly lower modularity ($Q = 0.1268$), confirming moderate but not dominant community structure. In contrast, the Girvan–Newman algorithm yielded a highly unbalanced partition (community sizes of 220, 11, 7 and 1) with a very low modularity ($Q = 0.006$). This reflects the algorithm's limitations in dense narrative networks, where edge betweenness does not isolate meaningful substructures.

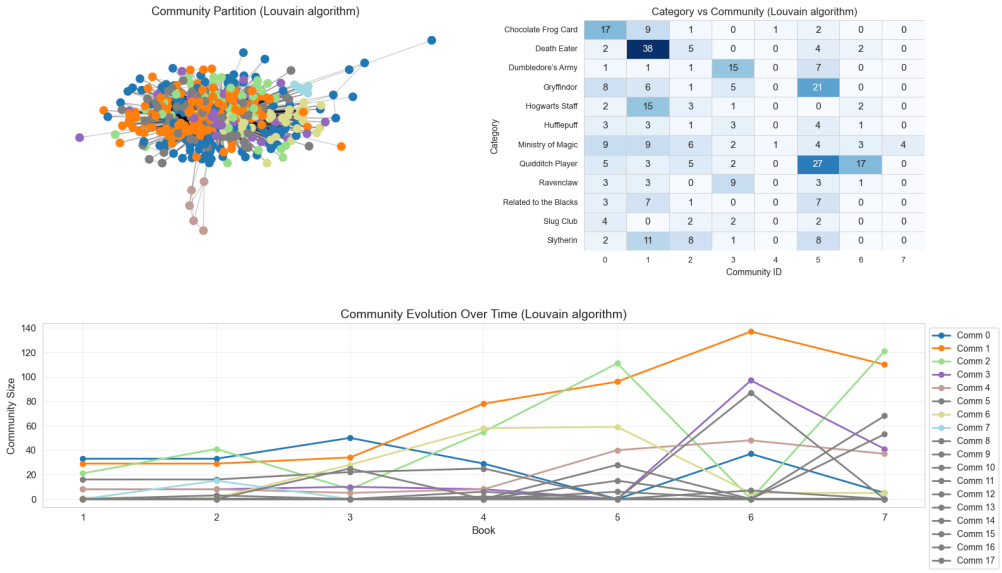


Figure 4.2: Louvain communities detected in the global Harry Potter network.

4.2.2 Overlapping communities

Non-overlapping methods assume that each character belongs to a single community, but this is unrealistic when characters participate in multiple groups. To capture this complexity, we applied the Clique Percolation Method (CPM).

Using a threshold of $k = 5$ ensures that communities correspond to tightly cohesive groups, in which every subset of five characters forms a fully connected subgraph. CPM with $k = 5$ discovered five overlapping communities in the global network, as summarised in Table 4.4.

CPM reveals a dominant large overlapping community of 381 characters, which reflects the highly interconnected core of the narrative: major characters co-occur in numerous scenes. Alongside this, CPM identifies four small but highly cohesive subgroups (sizes 8, 5, 5 and 5). They correspond to tightly bound character units with strong mutual interactions, such as small factions or families.

Unlike the non-overlapping methods, CPM allows characters to belong to multiple communities, providing a more nuanced view of how character groups overlap and interact within the story. The combination of CPM and Louvain therefore offers a complementary understanding of the network.

4.2.3 Category exploration by communities

The Louvain partition (left) displays a clearer organisational structure: *Death Eaters*, *Slytherins* and people *Related to the blacks* form a well-defined and concentrated block in Community 1, *Quidditch Players* from *Gryphindor* cluster strongly in Community 5, and the rest in 6. *Gryffindor* and *Ravenclaw* members appear primarily in Communities 0, 1, 3 and 5. Most of the *Chocolate frog cards* are grouped in communities 0 and 1, and the members of *dumbledore's Army* appear in community 3 (mainly from Gryffindor, Ravenclaw and

Community ID	Size	Num. Cliques	Density	Avg. Degree	Overlapping Nodes
0	5	1	1.000000	4.000000	3
1	5	1	1.000000	4.000000	3
2	381	2800	0.072952	27.721785	11
3	8	2	0.964286	6.750000	3
4	5	1	1.000000	4.000000	3

Table 4.4: Extended CPM community statistics for $k = 5$ in the global network. The table reports community sizes, number of 5-cliques, subgraph density, average internal degree, and number of nodes shared across multiple CPM communities.

Hufflepuf, which algins with the narrative in the books). This indicates that Louvain successfully recovers several functionally coherent groups, although some categories—such as the Hogwarts houses—are naturally spread across multiple communities.

In contrast, the CPM overlapping communities with $k = 5$ (right) reveal a markedly different structure: the vast majority of characters across most categories collapse into a single large overlapping community (Community 2), which is to be expected since the network is highly interconnected. Smaller CPM communities (IDs 0, 1, 3, and 4) capture only a few category elements each. Overall, Louvain separates broad functional groups more clearly, whereas CPM exposes the dense overlapping backbone of the narrative.

4.2.4 Community evolution

The community-trajectory plot shows that two dominant communities grow throughout the saga, reflecting the expansion and consolidation of the main narrative groups (Gryffindor, DA, Order of the Phoenix). From Books 6–7, communities merge into a few large clusters, indicating that the social structure becomes increasingly unified as the central conflict intensifies. Overall, the plot captures how the narrative shifts from local, disjoint storylines to a globally interconnected network in the final books.

5 Phenomenon Exploration

5.1 Role of Hubs

Hubs play an essential role in this network. They are highly connected nodes representing characters who co-occur with many others. Hence, they are indicators of narrative importance, since they have a high narrative presence, they are involved in multiple subplots, and they have relationships across diverse character groups. In addition, they are central for the spread of information and plot development. In this way, events or information associated with the main characters can quickly reach many others within the story.

5.2 Small-World Property

The expected average path length of a comparable random network [1] is

$$\frac{\ln N}{\ln\langle k\rangle} = \frac{\ln(440)}{\ln(24.745)} \approx \frac{6.086}{3.21} \approx 1.89.$$

Since the observed value is $\langle d\rangle = 2.02$, which is very close to the random expectation, the network exhibits the small–world structure, in which most characters are only a few steps apart. Therefore, this indicates a tighly interconnected character universe despite its size, where the overall narrative remains structurally unified rather than fragmented: information is spread quickly and major characters act as connectors.

5.3 Robustness

Studying the robustness of the network can be useful for the author or adaptation teams, since this highlights which characters are central to the narrative and need to be preserved to keep the story connected or what parts of the network become disconnected after the removal or temporary disappearance of a certain character. Therefore, it can be used as a tool for story coherence and audience comprehension.

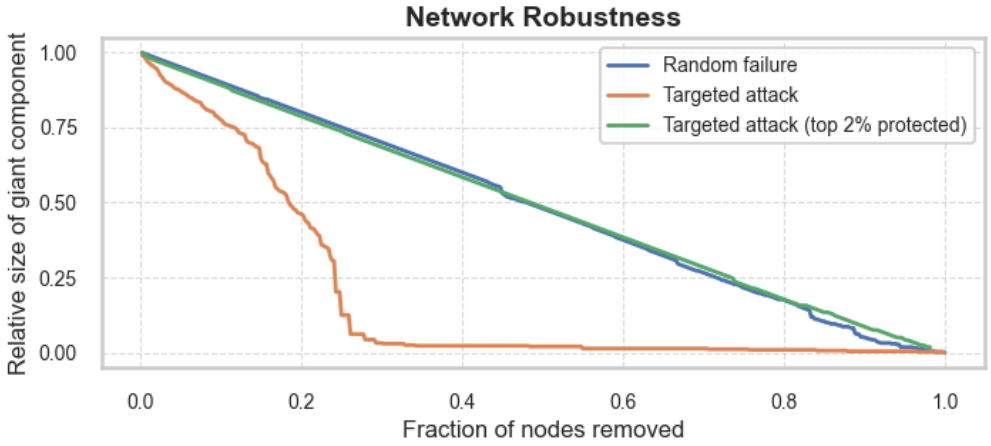


Figure 5.1: Simulated attacks on the global Harry Potter graph.

After simulating the attacks, we see that the network is robust to random failures-removing nodes at random produces only a gradual decrease in the size of the giant component-because most characters are minor and do not hold the network together. It is, on the other hand, extremely vulnerable to targeted attacks: removing high-degree hub characters rapidly fragments the network, with the size of the giant component dropping by half after removing only 20% of nodes. Protecting the top 2% of the hubs restores resilience, showing that a small set of key characters maintains structural cohesion.

5.4 Evolution

Since Harry Potter books follow chronological order, it is reasonable to study how the data evolve over time. We will use the chapters as our time unit. Next, we are going to discuss whether our network exhibits the elementary processes present in evolving network models [1] that could affect its topology.

Firstly, we are going to discuss **degree evolution of the nodes**. By looking at Fig. 5.3 we can observe that the biggest hubs (main characters) are introduced in the first chapters. Although we can observe the rich-get-richer effect, all nodes do not increase their degree at the same rate: some later introduced characters overcome other relevant previously introduced ones. Therefore, we have assumed that if there is preferential attachment in this network, not only does it depend on the degree of the node, but also on the **fitness**. We fit each node's degree by

$$\ln k_i(t) = \beta_i \ln t + B_i.$$

Then the fitness is

$$\eta_i = \frac{\beta_i - b}{a} \qquad a = \max_i \beta_i - \min_i \beta_i, \qquad b = \min_i \beta_i,$$

In Fig. 5.3 we can observe the empirical normalized fitness distribution. It is clear that the distribution is highly right-skewed due to the addition of more recent nodes. Nevertheless, we can still observe that nodes introduced in chronologically close regions exhibit distinct values. These value differences may reflect the narrative importance that the author unconsciously attributes to them.

Secondly, lets focus on **initial attractiveness**, which adds a constant to the preferential attachment function



Figure 5.2: Network Evolution Analysis.

that is proportional to the probability that a node acquires its first link in the next time step. However,after modeling that probability distribution (Fig. 5.3) we can observe that in all nodes this probability is set to one. This means that all nodes introduced into the network start with a degree $k > 0$. Therefore, there is no evidence of an inherent initial attractiveness mechanism since there are no isolated nodes that need to acquire links.

Thirdly, we have observed that there is a high presence of **internal links**, which are those links in a given chapter that join two characters that were introduced prior to that chapter. In Fig. 5.3 we can observe that the amount of internal links added is consistently bigger than the one of those links that join new characters with others among all chapters in the books. In the final graph, the internal links represent a 57.7% of the links added to the network. Since the author is (unconsciously) aware of the degree of the nodes she connects, the process behind is a Double Preferential Attachment model, where the likelihood that a new link emerges is proportional to the degree of the nodes it connects, rather than Random Attachment model. This aligns with the estimated degree exponent γ of our network and the fact that the network has a high heterogeneity, since Double Preferential Attachment lowers the theoretical Barabási–Albert degree exponent from 3 to 2.

With respect to **accelerated growth**, looking at Fig. 5.3 we can assume that there is no evidence of it because of two reasons. The first reason is that looking at the graph of the evolution of number of nodes vs. number of links, we can observe that its relation is approximately linear, with lower and higher rates in the initial and final global chapters, respectively. In addition, the impact of accelerated growth in the network would be increasing γ beyond 3 and making the network more homogeneous, which is not the case.

Finally, we have not considered that the processes of **aging** nor **node deletion** could influence this networks topology. In the context of the Harry Potter books, characters rarely disappear permanently once introduced:

major characters remain active throughout multiple chapters, and minor characters may reappear in later chapters even after initial appearances. As a result, there is no strong evidence that nodes age out or are removed from the network over time. Including aging or deletion mechanisms would likely reduce node degrees over time and fragment the network, which does not match the narrative dynamics of the book series.

5.5 Network Optimization

Since the Harry Potter network is shaped by storytelling rather than real dynamics, "optimization" must be understood narratively rather than functionally. If the goal is narrative coherence, then a highly centralized, scale-free network—with a few protagonists connecting most subplots—might be considered optimal. Tuning parameters such as the rate at which new characters are introduced or how often minor characters interact with major hubs would directly influence the clarity of the narrative. A lower attachment bias would create more independent subplots and a more fragmented network, while a stronger bias toward central hubs strengthens the unified storyline. In this sense, optimizing the network corresponds to choosing the level of narrative focus versus narrative diversity.

Looking back at Figure 5.3, we can observe how the number of nodes evolves over time: in the first chapters, only a few characters are introduced, but around the middle of the first book, the number of characters grows roughly linearly until the beginning of the last book, where the total number of characters seems to stabilize. Analyzing this kind of temporal patterns can serve as a useful template for other authors planning multi-book series, helping them balance narrative focus with narrative diversity.

6 Model comparisons and interpretation

In this section, we will describe how we have simulated different network models explained in [1] and contrasted them with our system's properties, aiming to find explainability in our network's topology. All models were created so that the number of nodes was the same as in the Harry Potter Network (440), and the number of links was close to the original value (5444). Results can be seen in Table 6.1 and Figure 6.1.

Network	L	$\langle k \rangle$	$k_{\mathbf{max}}$	$k_{\mathbf{min}}$	$\langle C \rangle$	$\langle d \rangle$	r	Connected	$\hat{\gamma}$
Harry Potter	5444	24.75	408	1	0.79	2.02	-0.33	True	2.10
Erdős–Rényi	5440	24.73	39	9	0.05	2.18	0.00	True	–
Watts–Strogatz	5280	24.00	30	19	0.37	2.48	0.01	True	–
Core–Periphery	5403	24.56	131	6	0.38	2.07	-0.29	True	–
Barabási–Albert	5037	22.90	106	8	0.10	2.22	-0.04	True	3.00
Bianconi–Barabási	5202	23.65	406	12	0.67	1.96	-0.44	True	2.11
D. P. Attachment	5189	23.59	138	5	0.37	2.28	-0.12	True	2.02

Table 6.1: Network statistics for generated models and the Harry Potter graph.

6.1 Erdős–Rényi

The probability p is chosen so that the expected number of edges matches the original graph:

$$\text{Expected edges} = p \cdot \binom{N}{2} = E \quad \Rightarrow \quad p = \frac{2E}{N(N-1)} = \frac{2 \cdot 5444}{440 \cdot 439} \approx 0.0564$$

Since $\langle k \rangle = 24.73 > lnN \approx 6.09$, the giant component has absorbed all nodes and the network is in the

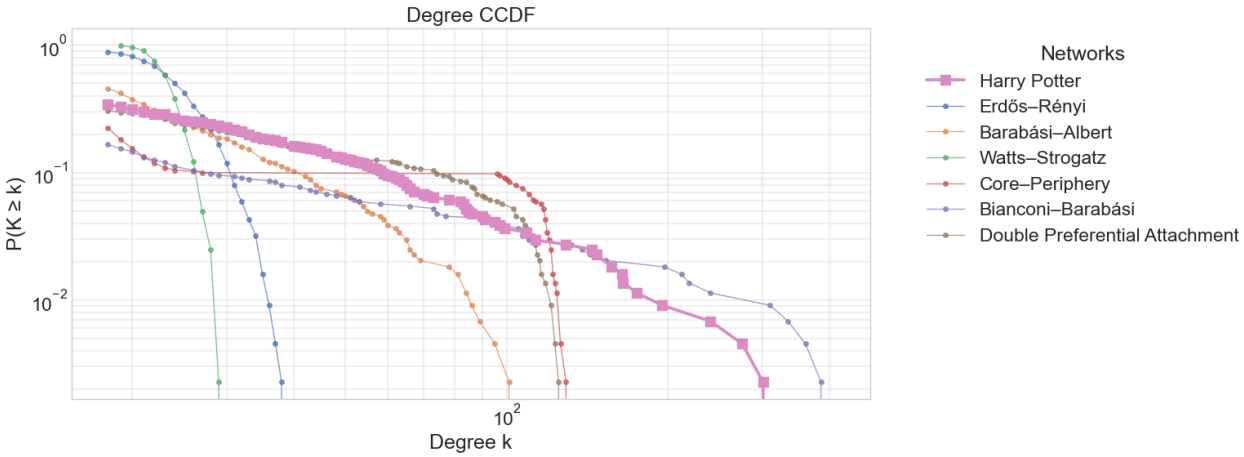


Figure 6.1: CCDF comparison of the networks.

Connected Regime, as so is the Harry Potter Network. In addition, as we showed before, the Harry Potter Network also exhibits the characteristic small-world property of Random Networks. However, the Erdős–Rényi model fails to capture the degree heterogeneity, since the ER model produces a roughly Poisson degree distribution whereas in the H.P. Network it follows a power-law with $\hat{\gamma} = 2.1.$, and therefore there are no hubs (main characters); the high local clustering observed in the social network; and it can't capture the correlations between the degree of connected nodes, due to the H.P. Network is disassortative ($r = -0.33$), whereas ER networks are typically neutral ($r \approx 0$).

6.2 Watts–Strogatz

The parameters of the model have been calculated as follows:

$$k = 2\left\lfloor \frac{\langle k \rangle}{2} \right\rfloor = 24, \quad p = \max(0.01, \min(0.9, 1 - \langle C \rangle)) \approx 0.21$$

Similarly to the Erdős–Rényi, this model succeeds in explaining the connectivity and small-world property of the studied network. Also, it is able to reproduce the moderate clustering ($\langle C \rangle = 0.37$), but fails to capture the scale free property (follows a narrow and bell-shaped distribution around $\langle k \rangle$), extreme clustering, degree correlations (it is nearly neutral), and the presence of low-degree nodes in the Harry Potter Network.

6.3 Core–Periphery Model

The core–periphery model [2] divides the N nodes into a set of core nodes and periphery nodes and assigns different connection probabilities depending on node type. This produces a network where the core is densely connected, the periphery is sparsely connected, and the core–periphery interactions reflect the original network's structure. We decided to test this model since it could resemble the distinction between main and secondary characters. Assuming that only 10% of the characters are truly relevant to the narrative, we calculated the parameters of the model:

$$N_{\text{core}} = \lfloor N \cdot f_{\text{core}} \rfloor = 44, \quad p_{\text{core}} = \frac{E_{\text{core}}}{\binom{N_{\text{core}}}{2}} \approx 0.316, \quad p_{\text{cp}} = \frac{E_{\text{cp}}}{N_{\text{core}} \cdot N_{\text{per}}} \approx 0.057, \quad p_{\text{pp}} = \frac{E_{\text{per}}}{\binom{N_{\text{per}}}{2}} \approx 0.052$$

The Core–Periphery model captures some global features of the Harry Potter Network, such as connectivity and small-world property, and reproduces the presence of high-degree core nodes and only the network's disassortativity ($r = -0.29$ vs -0.33. However, it underestimates the largest hubs ($k_{\max} = 131$ vs 408) and has a lower average clustering coefficient ($\langle C \rangle = 0.38$ vs 0.79). Moreover, the degree distribution exhibits a pronounced gap between low-degree periphery nodes and high-degree core nodes, reflecting the artificial core–periphery division rather than the smooth gradation of degrees observed in the real network.

6.4 Barabási–Albert

We calculated the initial parameters as follows:

$$m = \lfloor E/N \rfloor = 12, \quad N_0 = \lfloor N \cdot \text{main_char_percent} \rfloor = 22$$

The Barabási–Albert model reproduces the scale-free nature of the Harry Potter Network, generating hubs through preferential attachment and producing a power-law degree distribution with $\hat{\gamma} \approx 3.00$. However, it underestimates the largest hubs ($k_{\max} = 106$ vs 408) and overestimates the minimum degree ($k_{\min} = 8$ vs 1), resulting in a narrower degree spectrum. Additionally, the BA network exhibits very low clustering ($\langle C \rangle = 0.10$ vs 0.79) and nearly neutral assortativity ($r = -0.04$ vs -0.33), failing to capture the strong local cohesion and degree correlations present in the real network.

6.5 Bianconi–Barabási

In this case, we estimated the fitness distribution with the empirical character fitness distribution obtained in Fig. 5.3. The final parameters are the following:

$$m = \lfloor E/N \rfloor = 12, \quad \eta_i \sim \text{Beta}(0.1664, 3.232)$$

The Bianconi–Barabási model provides the best fit for the Harry Potter Network among the considered models. It successfully reproduces both global and local structural properties, including connectivity, small-world property, high clustering ($\langle C \rangle = 0.67$ vs 0.79), and disassortativity ($r = -0.44 \approx -0.33$). Importantly, the degree distribution closely matches that of the real network, capturing both the presence of extreme hubs ($k_{\max} = 406 \approx 408$) and the large number of low-degree nodes, and the estimated degree exponent ($\hat{\gamma} = 2.11 \approx 2.10$) is almost the same as in the real network. The only structural feature it does not fully explain is the exact abundance of the very low-degree nodes.

6.6 Double Preferential Attachment

We estimated the parameters in the following way:

$$N_0 = \text{round}(E_{\text{ext,t}}/N) = 5, \quad m = \text{round}(E_{\text{ext,r}}/(N - N_0)) = 5, \quad n = \text{round}(E_{\text{int,r}}/(N - N_0)) = 7$$

Finally, the Double Preferential Attachment (DPA) model reproduces many structural features of the Harry Potter Network, again including connectivity and small-world property, as well as moderate clustering ($\langle C \rangle = 0.37$). The model generates hubs with $k_{\max} = 138$, which captures the presence of high-degree nodes but underestimates the largest hubs of the real network ($k_{\max} = 408$). The degree distribution follows a heavy-tailed pattern with an estimated exponent $\hat{\gamma} = 2.02$, similar to the observed network. However, the model does not fully reproduce the very high clustering and strong disassortativity ($r = -0.12$ vs -0.33), and

low-degree nodes are somewhat overrepresented compared to the real network.

7 Conclusions and further work

Overall, we can conclude that the social network in the Harry Potter books can be modeled pretty accurately with a Bianconi–Barabási model, since it is able to explain its degree distribution, connectivity, small-world property, local clustering and degree correlation. This means that the underlying processes driving the formation of relationships in the series can be explained using growth and preferential attachment with fitness. We can therefore confirm that the social dynamics in the story are not random; they follow predictable patterns, where some characters become central hubs due to their narrative importance, while others occupy peripheral roles. This aligns with our theoretical expectations, since we expected the network to mimic a Barabási-Albert-based topology, due to the hypothesis that it followed a scale-free degree distribution.

These results demonstrate that network science can be applied to this domain, provinding real scientific findings beyond the mere curiosity of fans. It can serve as a practical tool for authors and film directors to gain a comprehensive overview of complex character networks, ensuring consistency and cohesion throughout the narrative. This, in turn, will lead to enhanced audience comprehension and improved subplot interconnections. In addition, such network-based analyses could prove useful when designing new stories or adaptations (by exploiting underutilized characters, or by highlighting narrative bottlenecks where the plot relies on a few central figures).

Regarding the project, the main aspect that could be improved is the graph generation method. The current technique of matching a static list of character names with a sliding window presents limitations, such as overlooking pronouns, name variations, merged identities or time awareness. Leveraging advanced techniques like large language models for entity recognition could address these issues, enabling more accurate character identification and network construction. Moreover, incorporating sentiment analysis and creating directed networks with relationship tags (e.g., friends, family, enemies, lovers) could provide deeper insights into narrative structure, character dynamics, and the evolution of the plot over time. These improvements would allow for a more precise model of the story's social network and a more realistic interpretation of the narrative.

Furthermore, this approach could be applied to different book series in order to investigate the existence of a universal pattern describing the social network structure in popular works of literature. Such a comparative analysis could reveal common underlying principles behind character interactions or highlight distinctive narrative strategies, offering new perspectives and tools for storytelling.

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